Role Uncertainty and Punishment Severity in Repeated Cheap Talk: Theory and Experiment

Jinyuqi Huang Wooyoung Lim

Department of Economics, HKUST

November 2026

Motivation

- Talk is cheap: Costless, non-binding, unverifiable.
- Sender talks, Receiver decides: To lie/trust or not depends on whether their incentives are aligned.
- When interests may conflict
  - One-shot: Uninformative communication (Crawford & Sobel, 1982).
  - Repeated: History-based strategies make truth-telling possible.

- We study two realistic features that shape repeated communication:
  1. **Role Uncertainty**: Sender/Receiver roles may vary.
  2. **Punishment Severity**: Payoff distribution in Grim via Nash-reversion.
- **Research question**: Individual effect of each feature on long-run communication, and their interaction effect.

Examples

Role uncertainty arise naturally in professional networks where Sender has **private information advantage** and Receiver has **decision power**:

- **Investment Syndication:** Firms share deal flows repeatedly. Today's initiator (Sender) may be tomorrow's participant (Receiver).
- **Academic Referral System:** Professors exchange evaluations of students. A letter writer (Sender) today reviews others' letters on a committee (Receiver) tomorrow.
- **The "Revolving Door":** Regulatory agencies and the private sector. Industry professionals (Senders) communicate compliance to regulators (Receivers), but frequently switch roles over their careers.

Punishment severity (how bad communication failure is) may depend on exogenous bargaining power etc.

The Model

Infinitely repeated cheap talk (cont. prob. δ) between players A and B.

Each round:

1. **Roles**: A is Sender, B is Receiver w.p. γ; opposite w.p. $(1-\gamma)$.
2. **State**: $\theta \in \{L, R\}$, equally likely, Sender's private information.
3. Sender sends message $m \in \{\text{“}L\text{”}, \text{“}R\text{”}\}$; Receiver picks action $a \in \{L, M, R\}$.
4. History is available: Past roles, states, messages, actions.

Payoffs (Sender, Receiver):
$b \in [0.5, 1)$

	L	M	R
$\theta = L$	$(1,1)$	$(1-b,\ b)$	$(0,0)$
$\theta = R$	$(1,0)$	$(1-b,\ b)$	$(0,1)$

Misaligned incentives:

- Sender always prefers action L; Receiver wants to match the state.

One-shot stage-Nash: Receiver ignores message, chooses M, payoff $(1-b, b)$.
Babbling equilibrium (BE) is inefficient, cannot achieve payoff $(1,1)$ in state L.

Equilibrium in Repeated Cheap Talk

Simple Truth-Telling Equilibrium (STE):

- Always tell the truth as Sender. Always follow message as Receiver.
- Deviation $\rightarrow$ permanent reversion to BE $\rightarrow$ $(1-b,\, b)$, **larger** b **means harsher punishment on Sender.**
- Simple for Receiver; gets full surplus; no incentives to deviate.
- Only Sender is tempted to deviate (lie in state R to get action L).
- STE is efficient, can achieve payoff $(1, 1)$ in state L.
- We use STE as theoretical benchmark due to its **strategic simplicity**.
- Other equilibria (e.g., Wilson & Vespa 2020) discussed in the paper.

Comparative Statics: STE

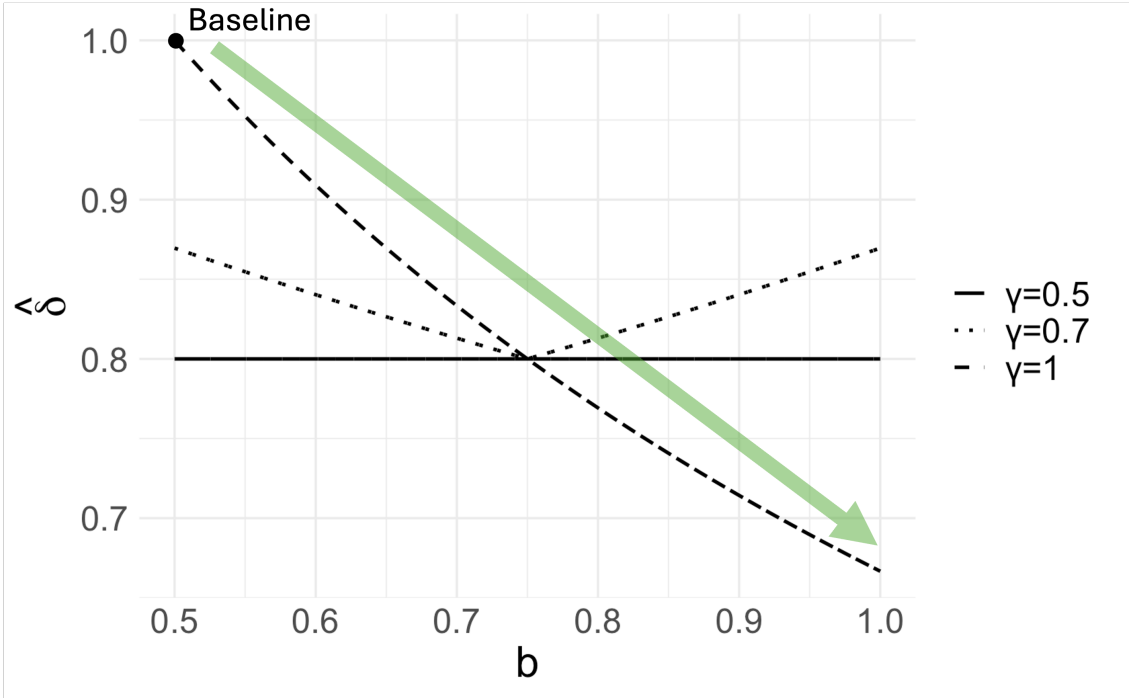
STE is sustainable when future cost $\geq$ current gain from lying.

- Current gain by lying is always 1.
- Future cost is $\frac{\delta}{1-\delta} C(\gamma, b)$.
- $\frac{\delta}{1-\delta} C \geq 1 \Rightarrow \delta \geq \hat{\delta}(\gamma, b)$. The smaller $\hat{\delta}$ is, the easier to support STE.

Comparative Statics w.r.t (γ, b):
Baseline: Fixed roles $(\gamma = 1)$ with mild punishment $(b = 0.5)$.

Comparative Statics: STE

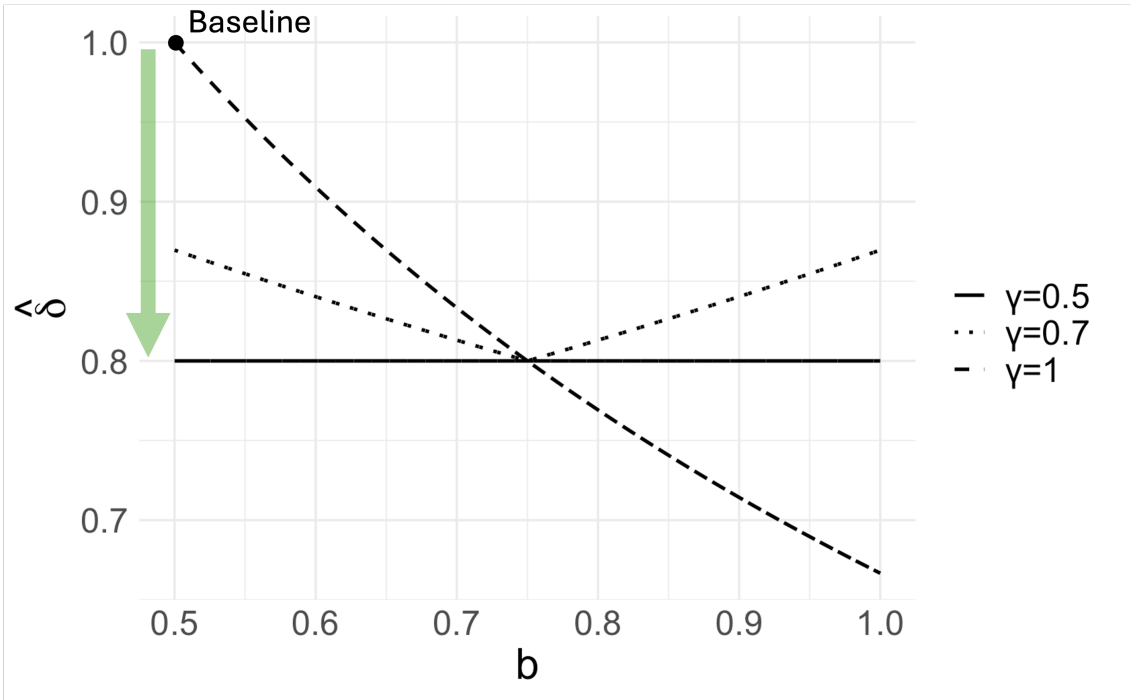
1. **Harsher punishment ($\uparrow b$) helps.**



Intuition: Sender's BE payoff is $1-b$.

Comparative Statics: STE

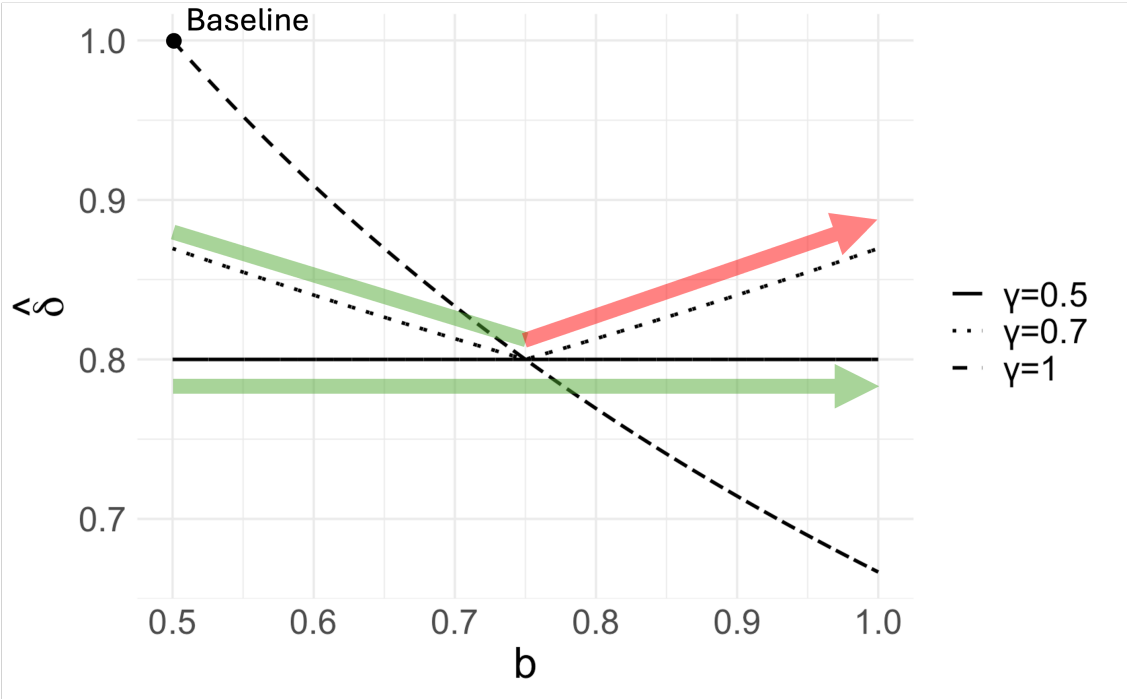
2. **More role uncertainty ($\downarrow \gamma$) helps.**



Intuition: Both players get to enjoy Receiver's information surplus, balancing incentives to cooperate.

Comparative Statics: STE

3. **Interaction: with $\gamma = 0.5$, no effect; with $\gamma > 0.5$, can backfire.**



Intuition: $\uparrow b$ creates two opposing forces.

- It raises Sender-self's cost but reduces Receiver-self's (BE payoff $b \uparrow$).
- Equally likely ($\gamma = 0.5$), effects on two selves cancel out.
- Unequally likely ($\gamma > 0.5$), Receiver-self dominates for player B who becomes harder to regulate.

Experimental Design & Predictions

Six Treatments: Role Uncertainty (γ) $\times$ Punishment (b)

	Fixed ($\gamma=1$)	Unequal ($\gamma=0.7$)	Equal ($\gamma=0.5$)
Mild ($b=0.5$)	**Baseline**	UR	ER
Harsh ($b=0.9$)	H	HUR	HER

- Implemented $\delta = 0.8$.
- Information transmission **efficiency** $\mathbb{P}(a = L|\theta = L)$.
- Truth-telling eqm. $\Rightarrow$ states revealed and followed, efficiency $= 1$.

Predicted efficiency relative to Baseline:

	H	UR	ER	HUR	HER
Prediction	$\uparrow$	$\approx$	$\uparrow$	$\approx$	$\uparrow$

Remark: Efficiency is *not* an identifying statistic for any TE. High efficiency can arise despite low truth if Receiver unconditionally trusts, and high honesty needs not translate into high efficiency if Receiver does not follow message "L". We complement efficiency with Sender and Receiver behaviors.

Implementation

Shared design (HKUST undergrads, no subject overlap)

	Wave 1	**Wave 2 (Primary)**
Time, Venue	2023 & 2025, via Zoom	Sep–Oct 2026, Lab
Sample Size	7 sess./treat., $N = 658$	4 sess./treat., $N = 360$
Supergames	7 supergames/session	**10** supergames/session
Termination ($\delta = 0.8$)	Random $\{2, .., 8\}$ rounds (procedural error)	**Block random termination** (5–15) (5 rounds/block)
Pre-registration	No	**Yes**
Consistency Across Waves	Highly consistent qualitative patterns across waves; only **HUR** shifts up in Wave 2; **Baseline**: more R after "R").	

- **Main analyses**: Exclusively **Wave 2** data.
- **Supplementary info**: Since Wave 1 has higher statistical power, we report its significance level **only when** Wave 2 has *same qualitative pattern but insignificant*.

View Replication Comparison

Results

	Mild ($b=0.5$)			**Harsh** ($b=0.9$)		
	Baseline	UR	ER	H	HUR	HER
Efficiency $\mathbb{P}(a = L \mid \theta = L)$	0.47	**0.57****	**0.58**	0.48	0.49	0.42
Truth $\mathbb{P}(m = \text{“}R\text{”} \mid \theta = R)$	0.40	0.40	0.35	**0.58****	**0.52***	0.48
Trust $\mathbb{P}(a = L \mid m = \text{“}L\text{”})$	0.47	**0.54****	**0.57**	0.46	0.45	0.40*

Significance vs. Baseline (two-way clustered SEs): $^{*}p < 0.1$,** $p < 0.05$,*** $p < 0.01$ in Wave 2; ** denotes significance in Wave 1 (same direction).

Three striking findings:

1. H: Harsh punishment promotes Sender truth but not Receiver trust.
2. UR/ER: Role uncertainty triggers trust despite unchanged truth.
3. HUR/HER: Interaction cannot elicit trust despite higher honesty.

They challenge two implicit assumptions in STE-based predictions.

Safety Trap: Harsh Punishment Treatments

Expected: ↑ punishment → ↑ truth → ↑ trust → ↑ efficiency.
Actual: H, HUR and HER → ↑ truth → ∼ or ↓ trust → ∼ efficiency.

Why? Same parameter that punishes Sender harder makes Receiver's action M more attractive (guarantees a safe **0.9** regardless).

- Receivers default to M instead of trying to trust.
- Role uncertainty may or may not help: Fail to integrate two roles in the long run. Selfish choice feel less unfair.

Evidence: $\mathbb{P}(a = M \mid m = \text{“}L\text{”})$ is much higher in harsh treatments:

	Baseline	UR	ER	H	HUR	HER
$\mathbb{P}(a = M \mid m = \text{“}L\text{”})$	0.31	0.36	0.33	**0.52*****	**0.52*****	**0.58*****

In HUR and HER, first M-to-“L” occur in round 1 by 10% more than mild punishment treatments. Trapped by safety-seeking in the beginning, not punitive.

Blind Trust: Random Role Treatments

Expected: Role uncertainty $\rightarrow \uparrow$ truth $\rightarrow \uparrow$ trust $\rightarrow \uparrow$ efficiency.
Actual: UR and ER $\rightarrow \sim$ truth, but $\uparrow$ trust $\rightarrow \uparrow$ efficiency.

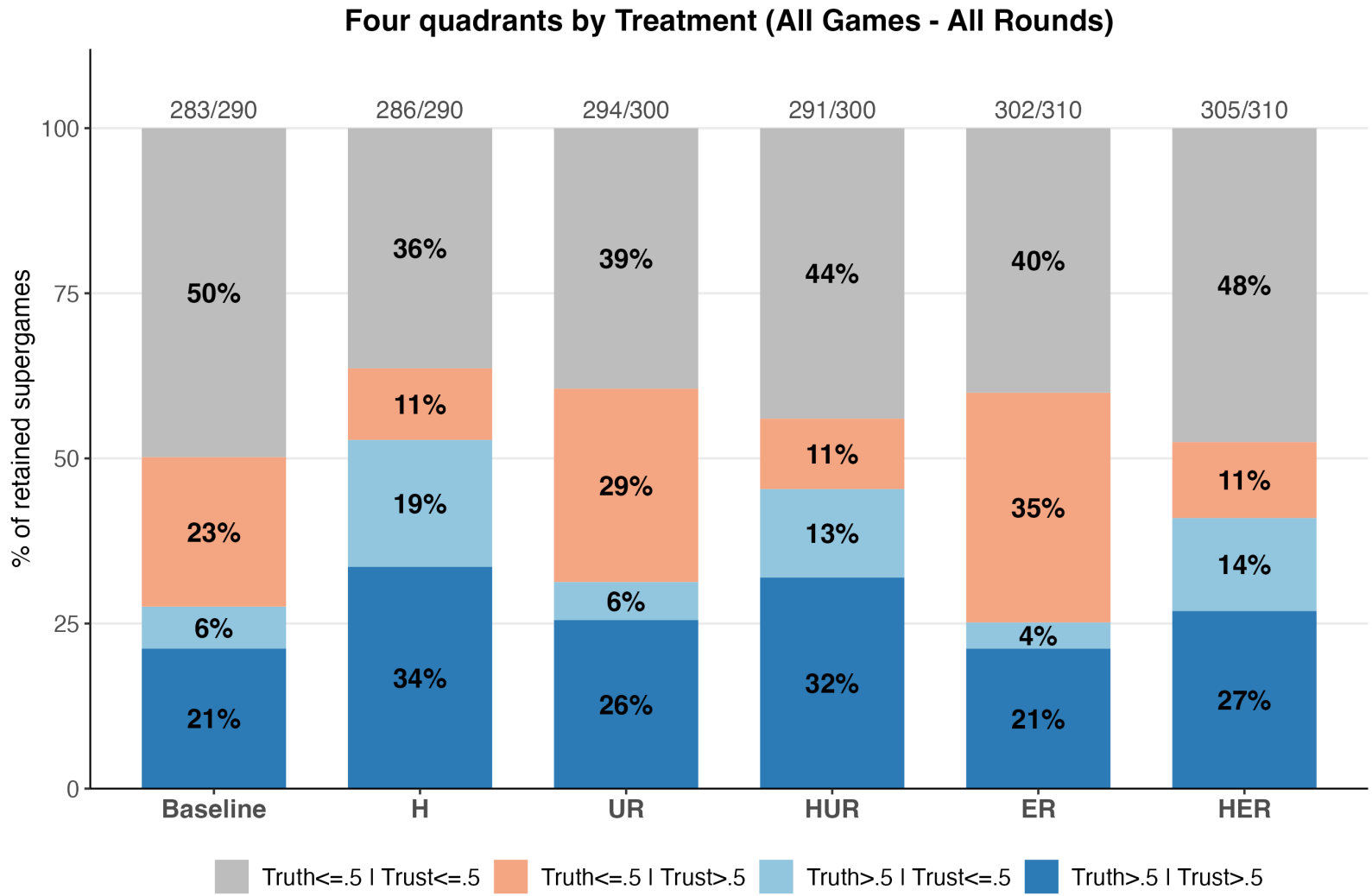
Blind Trust Equilibrium (BTE):

- Receiver *always* trusts "*L*". Players take turns to get *L* as Sender.
- Efficiency $\mathbb{P}(a = L \mid \theta = L) = 1$, but **no info. is actually transmitted**.

Why does BTE emerge in UR & ER but not elsewhere?

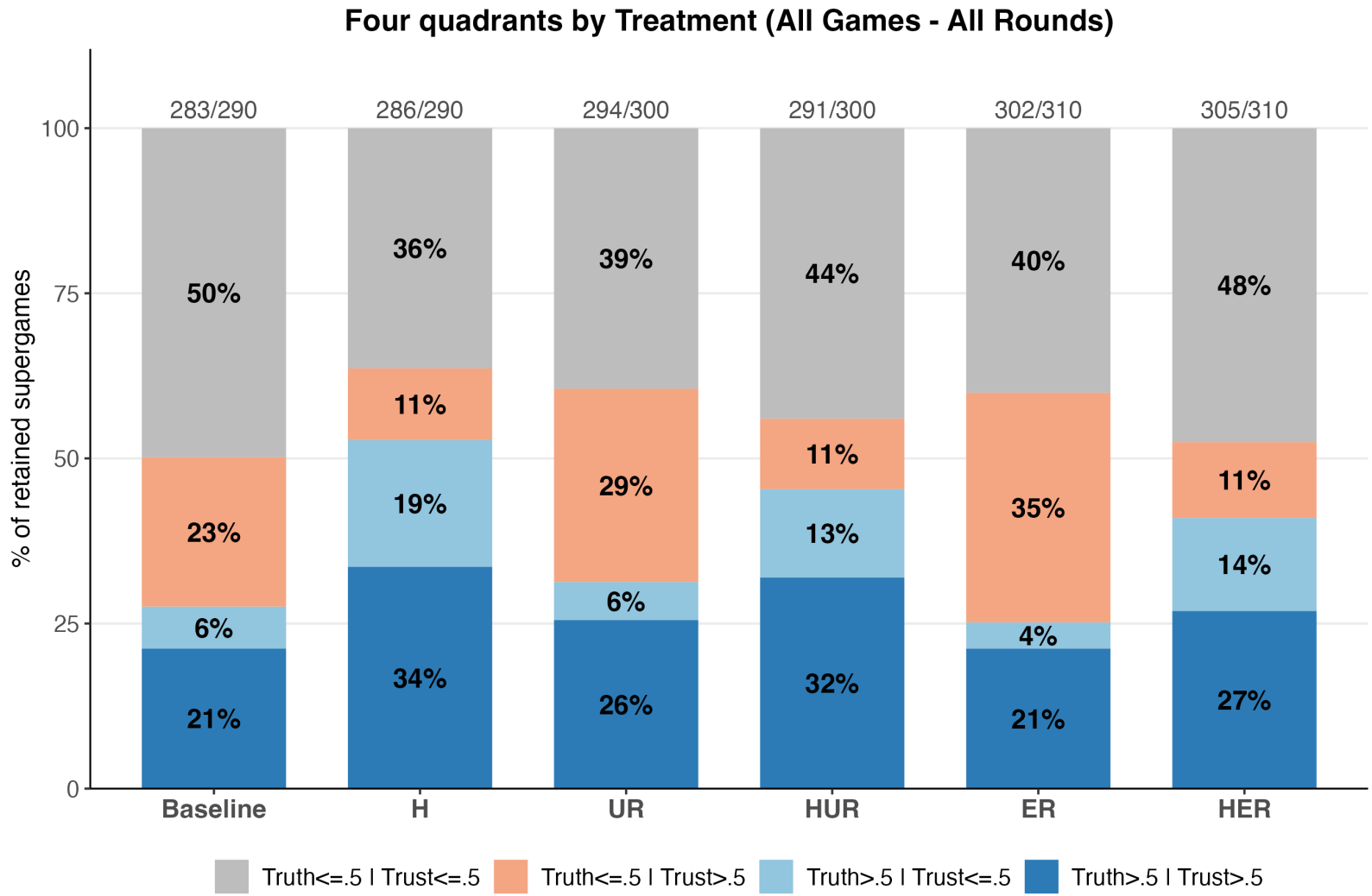
- **Trivial** $\hat{\delta} = 0$ for UR and ER.
  - Harsh + role uncertainty: $\hat{\delta} = 0.62$ in HER; impossible in HUR.
  - Fixed roles: BTE is undefined (Baseline, H).
- **Strategically simple**: Sender needn't resist lying; Receiver needn't assess credibility.

Supergame-Level Evidence: Safety Trap



- H, HUR, HER have larger fraction in high-truth but low-trust.

Supergame-Level Evidence: Blind Trust



- UR, ER have larger fraction in low-truth but high-trust.

Takeaways & Next Steps

We study how role uncertainty and punishment severity jointly shape communication in repeated cheap talk.

- Theory (STE): Each feature alone helps, while interaction may hurt.
- Experiment data reveal different patterns:
  - **Safety trap:** Harsh punishment improves truth but fails to elicit trust.
  - **Blind trust:** Role uncertainty leads to efficient but untruthful outcome.

Next steps:

1. **Break the Safety Trap**: Lower the Sender's stage-Nash payoff while holding the Receiver's stage-Nash payoff fixed.
2. **Penalize uninformed decisions**: BTE may be less favorable than STE if information has intrinsic value. We can reduce Sender's payoff for her preferred action if Receiver is misled, capturing the welfare loss of misallocating the resource.

Literature

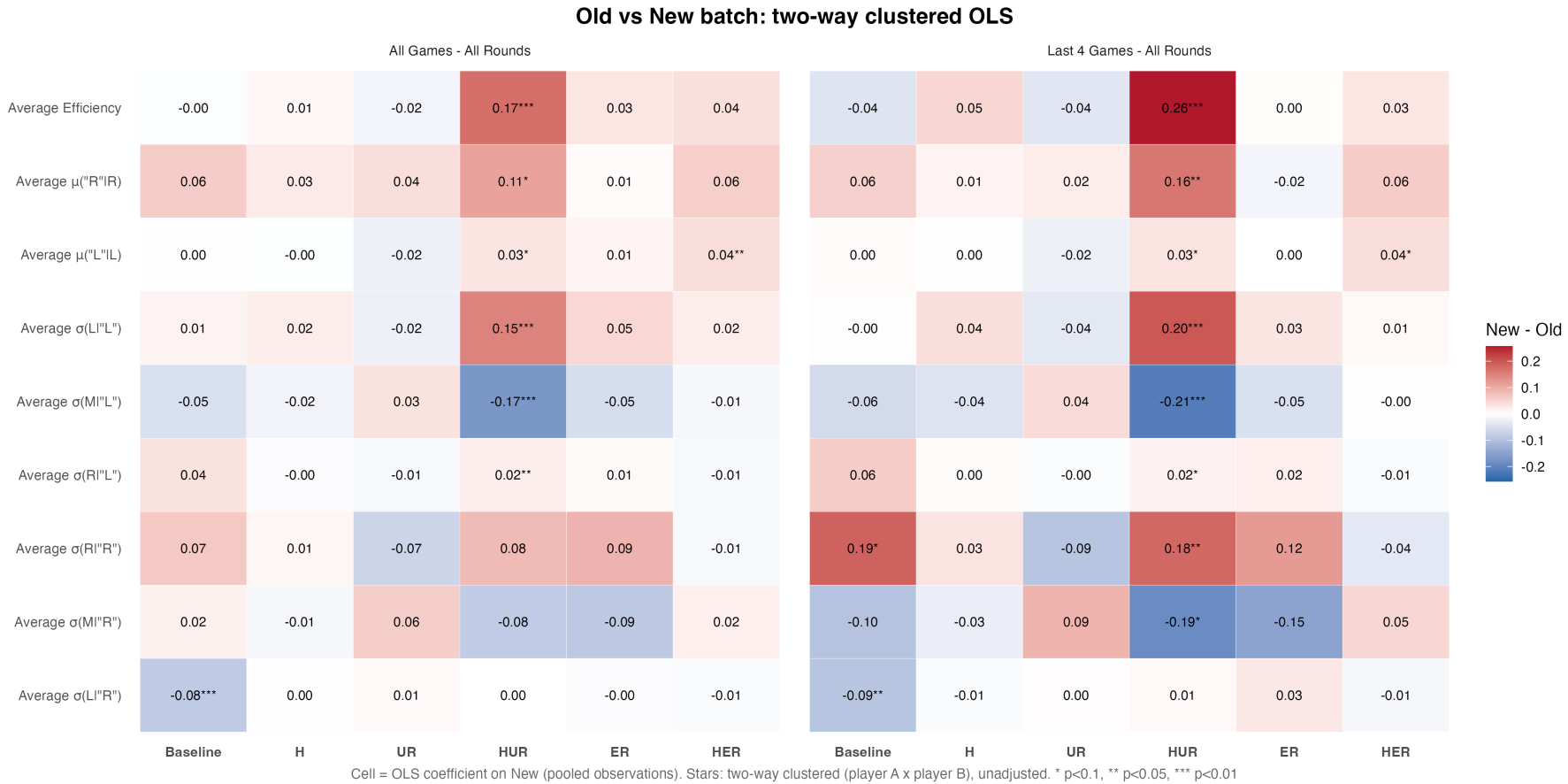
- **Facilitating communication in cheap talk:** Crawford and Sobel (1982); Blume, Lai and Lim (2020); Golosov et al. (2014); **Wilson and Vespa (2020)**; Kolotilin and Li (2021); Kuvalekar, Lipnowski and Ramos (2022); Best and Quigley (2024).
- **Role switching in games:** Classen (2005); Iriberri and Rey-Biel (2011); Ben-Ner et al. (2004); Herne, Lappalainen and Kestila-Kekkonen (2013); Burks, Carpenter and Verhoogen (2003); Johnson and Mislin (2011).

Thank You!

Questions or comments?

jhuangde@connect.ust.hk
wooyoung@ust.hk

Replication Consistency: W2 (Lab) minus W1 (Online)



Cell = OLS coefficient on New (pooled observations). Stars: two-way clustered (player A x player B), unadjusted. * p<0.1, ** p<0.05, *** p<0.01

- **Overall**: Highly consistent; qualitative treatment comparisons are preserved.
- **HUR**: Higher efficiency, driven by more truth and trust.
- **Baseline**: only more R less L to "R".
- **Why**? Better focus in lab; longer supergame horizons (5–15) encourage cooperation (upward shifts in H, ER, HER).

◂ Back